R2YOLOX: A Lightweight Refined Anchor-Free Rotated Detector for Object Detection in Aerial Images

Fei Liu, Renwen Chen, Junyi Zhang, Kailing Xing, Hao Liu, and Jinchang Qin

***Abstract*— Existing anchor-based rotated object detection methods have achieved some amazing results, but these methods require some manual preset anchors, which not only introduce additional hyperparameters but also introduce extra computational burdens. Due to the above drawbacks, anchor-free methods have been rapidly developed in recent years. However, the existing high-performance anchor-free rotated object detection methods are relatively complex and the inference speed is also slow. And you only look once detector (YOLO) series models not only maintain high-efficiency inference but also keep competitive performance detection performance in the general object detection tasks. Hence, we propose an anchor-free rotated detector based on the exceeding you only look once detector (YOLOX) method for object detection in aerial images. Our methods consist of two improvements: a refined rotated module (RRM) and a new assigner method which is called the Gaussian distribution sampling optimal transport assignment (GSOTA) method. The RRM can align features and get more useful priors for final detector heads. The GSOTA uses Gaussian distribution to model the oriented bounding box (OBB) first, and a Gaussian center sampling (GCS) method with maximum classification center mean (MCCM) is proposed to simplify the label Assignment optimal transport (OT) problem, finally using an improved dynamic top-*k* strategy to get an approximate solution. Extensive experiments demonstrate that our models can achieve competitive performance in several challenging aerial object detection datasets while keeping the best efficiency. Our R2YOLOX-X model achieves 79.33%, 97.4%, 97.7%, and 92.5% mean of Average Precision (mAP) on the dataset for object detection in aerial image (DOTA), HRSC2016, University of Chinese Academy of Science (UCAS)-dataset of object detection in aerial image (AOD), and FGSD2021, respectively, while R2YOLOX-S can reach the fastest 58.2 FPS when inferencing on aerial datasets and R2YOLOX-L gets the best speed-accuracy trade-off.**

***Index Terms*— Aerial image, anchor-free, deep learning, rotated object detection.**

I. Introduction

AERIAL images are typically taken from a bird's-eye view, which implies that objects in aerial images are always arbitrarily oriented. To achieve accurate object location in the aerial object detection tasks, the oriented bounding box (OBB) was used to represent objects [1], [2], [3], [4]. Moreover, the highly complex backgrounds and variant appearances of objects further increase the difficulty of object detection in aerial images (ODAIs) compared with common object detection. Therefore, ODAIs become an emerging research direction and has attracted more and more attention in recent years. With the framework of deep convolutional neural networks, ODAI has made significant progress in the past few years [1], [5], [6], [7], [8], [9], [10], [11], [12]. The most state-of-the-art (SOTA) aerial object detectors [7], [10], [12], [13] rely on the two-stage detection paradigm [14], which consist of two parts: a region proposal network (RPN) and an region-convolution neural network (R-CNN) detection head. Usually, RPN is a lightweight head and generates some high-quality region of interests (RoIs) through horizontal anchors, then some RoI pooling operators are used to extract more accurate features from RoIs. The R-CNN head is adopted to regress the location of objects and classify them into different categories. However, using horizontal bounding boxes (HBBs) to represent objects or using horizontal RoIs in ODAI often results in severe misalignments between predicted bounding boxes and rotated objects [1]. Hence, recent open-source datasets [1], [6] provide annotations in the form of OBB, and Ding et al. [7] proposed the RoI transformer to transform horizontal RoIs into rotated RoIs that avoid a large number of anchors with different angles [15], but this method still requires massive computations and memory footprint. In contrast with two-stage detectors, one-stage detectors regress the bounding boxes and classify them directly with regular and dense sampling anchors. This architecture enjoys high computational efficiency but often lags behind in accuracy [1]. Recent work [11] reports that the performance disadvantage of one-stage detectors mainly comes from feature misalignment. This misalignment is due to the convolution features from thebackbone network being axis-aligned with a fixed receptive field, while objects in aerial images are distributed with arbitrary orientations and variant appearances, hence the corresponding feature of the

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axis-aligned grids is hard to represent the whole rotated object to some extent. Therefore, Han et al. [11] propose a method named AlignConv to address one-stage feature misalignment issues and got a significant improvement.

However, the above-mentioned aerial object detection methods all belong to the anchor-based object detector, and these methods require manual preset anchors, which not only introduce additional hyperparameters and calculations but also directly affect the efficiency of the model. The anchor-free methods remove the preset anchors and reduce the prior information, which makes them more adaptable than the anchor-based methods in common object detection tasks. Therefore, for deploying the advanced aerial object detection algorithm on the unmanned aerial vehicle (UAV) or satellite in the real world, how to modify the anchor-free method to fit the ODAI tasks has become an emerging research direction.

Recent works [16], [17], [18], based on the anchor-free paradigm have achieved some progress on ODAI, but the above methods all use complex transformation functions to rerepresent the original OBB, which are difficult to be deployed into the actual embedded device. In addition, the method [19] proposed an anchor-free RPN method but it belongs to the two-stage detector which has a complex architecture and low inference efficiency.

Hence, in this article, we deliver a refined anchor-free rotated detection method called R2YOLOX, which is based on the recent anchor-free method exceeding you only look once detector (YOLOX) [20]. We do not use any complex representation transformation methods, and through one center key points directly regress the OBB of objects and classify them into different categories. Therefore, to the best of our knowledge, our proposed method achieves the best balance of performance and accuracy.

A series of experiments on dataset for object detection in aerial image (DOTA), HRSC2016, University of Chinese Academy of Science (UCAS)-dataset of object detection in aerial image (AOD), and FGSD2021 datasets verify the effectiveness of our method. In summary, our main contributions include the following.

1) Our proposed Gaussian distribution sampling optimal transport assignment (GSOTA) label assignment method uses Gaussian distribution characteristics to modify the OT assignment (OTA) method. Our GSOTA is more appropriate with oriented objects in aerial images and gains remarkable performance improvement.
2) To overcome the inherent feature misalignment in one-stage detectors, the AlignConv technique [11] is used in our refined rotated module (RRM). By using RRM and GSOTA in baseline simultaneously, we propose a one-stage anchor-free rotated detector R2YOLOX, which not only has the best speed of inference (i.e., the fastest can reach 58.2 FPS) but also achieves competitive detection accuracy [i.e., the best can reach 79.33 mean of Average Precision (mAP)].
3) We conduct extensive experiments, and the results demonstrate that our method achieves competitive performance when compared with other anchor-based methods, and compared with other anchor-free methods our method achieves SOTA. Our method greatly reduces the gap between anchor-free and anchor-based methods and has the best speed-accuracy trade-off.

II. Related Work

In this section, we first review the related methods of rotated object detection (also called oriented object detection) in aerial images. Then, we briefly introduce the recent progress of anchor-free object detection methods in aerial images. Finally, we study the label assignment strategy for training the object detectors.

A. Rotated Object Detection in Aerial Images

In the past decade, remarkable progresses [14], [21], [22], [23], [24], [25], [26] have been made in the field of object detection. As an extended branch of object detection, oriented object detection [5], [7], [10], [11], [12], [13], [15], [16] has received extensive attention. It can be widely used in management, precision agriculture, national defense, emergency rescue, and disaster relief. A natural solution to oriented object is to set rotated anchors [6], [15], such as rotated RPN [15], in which the anchors with different angles, scales, and aspect ratios are placed on each location. These densely rotated anchors lead to extensive computations and memory footprints. To decrease a large number of rotated anchors and reduce the mismatch between features and objects, Ding et al [7] proposed the RoI transformer that learns rotated RoIs from horizontal RoIs produced by RPN. This method greatly boosts the detection accuracy of oriented objects. However, it makes the network heavy and complex because it involves fully connected layers and RoI alignment operation during the learning of rotated RoIs. Xu et al. [10] proposed a new oriented object representation, termed gliding vertexes. It achieves oriented object detection by learning four vertex gliding offsets on the regression branch of the faster R-CNN head [14]. However, these two methods both adopt horizontal RoIs to perform classification and OBB regression. They still suffer from severe misalignment between objects and features.

In addition, some work [11], [27], [28] have explored one-stage oriented object detection frameworks, which output object classes and OBBs without region proposal generation and RoI alignment operation. For example, Yang et al. [27] proposed a refined one-stage oriented object detector, which involves two key improvements, including feature refinement and progressive regression, to address the problem of feature misalignment. Han et al. [11] proposed a single-shot alignment network (S^2ANet) for oriented object detection, which aims at alleviating the inconsistency between the classification score and location accuracy via deep feature alignment. Ming et al. [28] designed a new label assignment method for rotated object detection that could assign the positive and negative anchors dynamically. In addition, above-mentioned methods are all based on the RetinaNet [22] framework.

B. Recent Anchor-Free Object Detector in Aerial Images

In the past two years, anchor-free detectors [20], [24], [25], [29], [30] have developed rapidly in the general object detection tasks. These works are mostly one-stage architecture and the results have shown that the performance of anchor-free detectors can be competitive with anchor-based detectors. Anchor-free mechanism significantly reduces the number of

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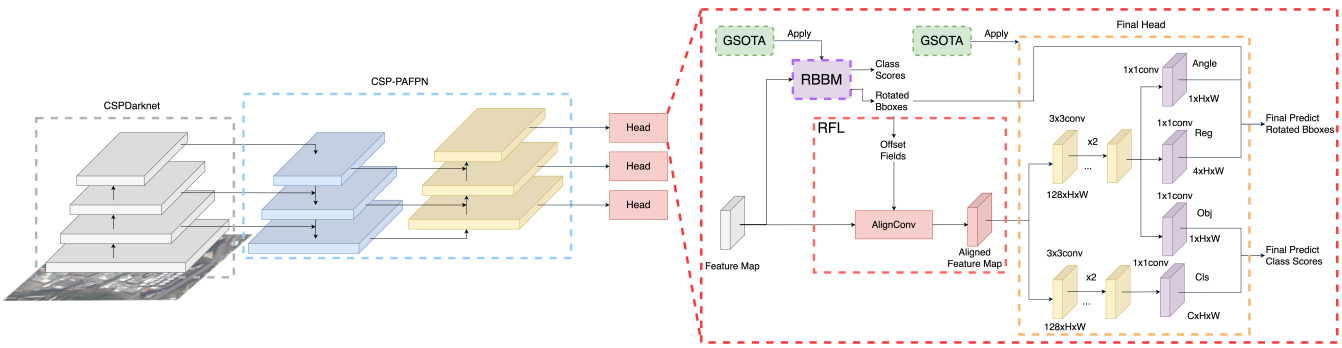

Fig. 1. Overall architecture of our proposed R2YOLOX.

design parameters that need heuristic tuning and many tricks involved for good performance, making the detector, especially its training and decoding phase, considerably simpler [20], [24]. But it is still a challenge to use anchor-free methods in the aerial images object detection task. Because the targets in aerial images are more difficult than common objects.

Recently, Pan et al. [16] devised a dynamic refinement network (DRN) for oriented object detection based on the anchor-free object detection method CenterNet [30]. He et al. [31] leveraged attention mechanisms [attention refinement one-stage anchor-free detector (AROA)] to refine the performance of remote sensing object detection in a one-stage anchor-free network framework. Cheng et al. [19] proposed an anchor-free oriented proposal generator (AOPG) to replace the horizontal boxes-related RPN in the faster R-CNN detector and got improved performance. Li and Zhu [18] proposed an effective anchor-free method named Oriented RepPoints, which uses an adaptive point set as a fine-grained representation to capture the semantic and geometric features of an oriented object. Cui et al. [32] focused on rotated ship (SH) detection in optical remote sensing images, an anchor-free SH detection method rotated ship keypoint detection network (SKNet) [32] is proposed, which mainly combines the Orthogonal Pooling, Soft Rotate non-maximum suppression (NMS), and CenterNet's key point estimation detection method; in addition, they also proposed feature attention transfer for ship keypoint detector (SKFat) [33], which used the transfer attention mechanism and inclusion-exclusion learning supervised method to enhance the multiscale features of the anchor-free method, and has achieved significant performance improvement for the SH detection task. Zhang et al. [34] proposed center-head point ship detector (CHPDet) [34] based on the CenterNet method, which used key point estimation to regress the center point and head point of SHs, and when ground sample distance (GSD) is known, it further used SH size as prior knowledge to filter out redundant results.

Mostly, anchor-free methods directly predict the bounding box of the object, hence these methods usually use the intersection over union (IoU) base loss function to regress the object location in the general object detection tasks. But rotational IoU losses are indifferentiable and have boundary discontinuity and square-like problems when using the OBB representation, these problems hinder the use of anchor-free methods. Therefore, recently researchers have been committed to the development of differentiable approximate rotated IoU loss. Based on Gaussian Wasserstein distance (GWD)

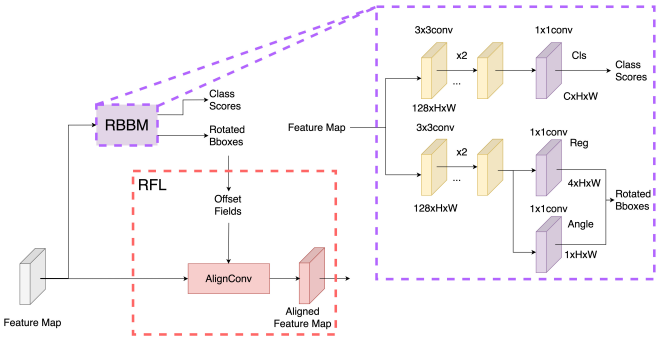

Fig. 2. Architecture of RRM. For each input feature map of the FPN, first use RBBM to get coarse rotated bounding boxes and then through RFL refined the features to align with the coarse rotated bounding boxes. Finally, output the aligned feature map and refined priors.

and Kullback-Leibler divergence (KLD), Yang et al. designed GWD [35] and KLD [36] regression loss, respectively. Using these methods in the anchor-free method fully convolutional one-stage object detector (FCOS) [24] can get performance improvements on rotated object detection tasks. Chen et al. [37] used the total number of pixels covered by OBB to replace the calculated actual OBB areas, derived an approximate rotated IoU loss named pixels-intersection over union (PIoU). Then, experiment results show that PIoU can improve the performance of the CenterNet [30] on rotated detection tasks.

C. Label Assignment Method

Most detectors select positive and negative samples through a specific strategy, which is called the label assignment method. The commonly used label assignment method is to divide the positive and negative samples by the IoU between the anchor box and the ground-truth (gt) bounding box, which is called the Max-IoU method. For instance, RetinaNet selects positive samples with IoU values higher than 0.5, and those whose IoU values are lower than 0.4 are considered to be negative samples. The Max-IoU method selects the samples by measuring the spatial relationship between the preset anchor and the ground truth. It has strong spatial constraints on the candidate samples.

In recent years, the label assignment strategy based on the relationship of central distances has been proposed with the emergence of the anchor-free method. FCOS [24] divides positive and negative samples by measuring the spatial-scale limitation and whether the anchor point falls into the ground truth or not. Adaptive training sample selection (ATSS) [38]

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analyzes the main differences between FCOS and RetinaNet, a label assignment method based on the center distance constraints is proposed, detailly ATSS selects the candidate samples through the center distance relationship and determines the positive threshold from the statistical information. OTA [39] analyzes the label assignment from a global perspective and formulates the assigning procedure as an optimal transport (OT) problem, producing the SOTA performance among the current assigning strategies [24], [38], [40]. However, in practice, researchers found solving OT problems via the Sinkhorn-Knopp algorithm brings 25% extra training time, which is quite expensive for the computation. Therefore, the YOLOX [20] serials model uses a simplified OTA method named SimOTA to get an approximate solution. But this method is designed for generic object detection tasks. It does not work well for the rotated object detection situation. For rotating object detection, the ground truth has an arbitrary angle, and the angle change will greatly affect the IoU. Therefore, the Max-IoU method is not conducive to selecting a large number of high-quality candidate samples in the rotation detection task. The above-mentioned anchor-free label assignment strategies do not consider the angle information, so they are less restrictive sample selective strategies, which is not conducive to further assigning high-quality samples on aerial image object detection tasks.

III. Method

This section first modifies YOLOX architecture for rotated object detection and selects it as the baseline. Then, we detail the RRM and GSOTA methods. Finally, we describe the details of our proposed R2YOLOX detectors, and the whole model architecture is shown in Fig. 1.

A. Rotated YOLOX as Baseline

We choose a one-stage anchor-free detector: YOLOX as our baseline. It consists of a backbone network and a decoupled head. The backbone network is mainly based on DarkNet53 including modified cross stage partial network (CSPNet), sigmoid linear unit (SiLU) activation, and the PAFPN neck. It also follows the YOLOv5 scaling rule to produce YOLOX-S, YOLOX-L, and YOLOX-X models. The decoupled head has two task-specific subnetworks for classification and localization, this structure is widely used in most one-stage and two-stage detectors [22], [41]. Each subnetwork consists of two stacked fully convolution layers and a prediction convolutional network.

Note that YOLOX is designed for generic object detection, the network predicts an HBB to indicate the object, which can be represented as follows:

$$(x, y, w, h) \tag{1}$$

where x, y are the center coordinates of the bounding boxes, and w, h are the width and height of the bounding boxes. To be compatible with oriented object detection, we just add a rotated angle to the regression output of the YOLOX head, represented by θ Therefore, the regression prediction of rotated YOLOX models is the OBB, represented as

$$(x, y, w, h, \theta) \tag{2}$$

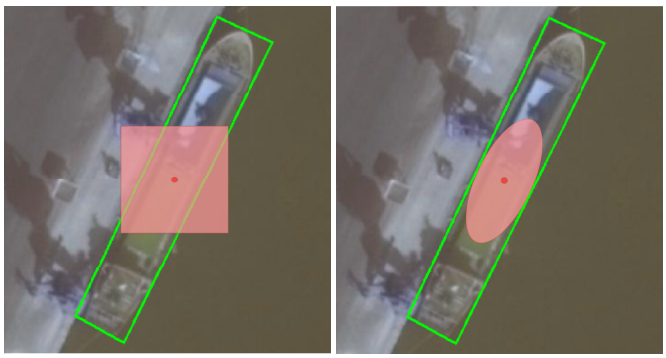

Fig. 3. Comparison of the difference between traditional center sampling methods and our proposed method. (a) Traditional square center sampling range on the OBB. (b) We propose the GCS region on the OBB.

where $\theta \in [-(\pi/2), (\pi/2)]$ denote the clockwise rotated angle from the image coordinate system position direction of x to the bounding box's relatively coordinate system position direction of x, and we use the long edge definition format [35] in which the width w is always larger than the height h. Due to our computation source limit and alignment with the experiment environment with other related work, we made the following modifications on the baseline.

1) We use the cross stage partial darknet (CSPDarkNet)-S model as the pretrained backbone, which is from the open MultiMedia Lab detection toolbox and benchmark (MMDetection) framework trained on the Microsoft Common Objects in Context dataset (COCO).
2) Because the IoU loss cannot be directly used on rotated detection tasks [35], we use the KLD loss to replace the original YOLOX regression loss.
3) The experiment shows that the loss of the model almost converges at 40 epochs when using the pretrained backbone coming from the COCO. Therefore, we train 40 epochs to get the model, instead of training 300 epochs from the scratch.

In the end, all other training configs keep unchanged with the original YOLOX method.

B. Refined Rotated Module (RRM)

This section introduces the RRM that consists of a rotated bounding box module (RBBM) and refined feature layers (RFLs), which are used to refine the rotating feature and get more precise priors, and this module structure is illustrated in Fig. 2.

1) Rotated Bounding Box Module: The RBBM is a lightweight network with two parallel branches, a classification branch and a regression branch. Each branch consists of a 2-layer 3×3 convolution layer and a prediction output layer. For the classification branch, the output layer is a 1×1 convolution layer and the convolution channels are equal to the class categories.

Then, for the regression branch, the output layer consists of two 1×1 convolution layers, The first one is responsible for predicting a distance vector (l, t, r, b), which represents the distance from the center of positive sample priors to the left,

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top, right, and bottom boundaries of the ground-truth bounding box likes FCOS [24], and the other layer is responsible for predicting rotated angle θ. The structure of this module is shown in Fig. 2.

2) Refined Feature Layers: In this section, we apply a feature alignment technique called AlignConv [11] to align the input features with RBBM decoded output coarse rotated bounding boxes. AlignConv is a deformable convolution layer, different from normal deformable convolution AlignConv uses RBBM decoded outputs as the offset field instead of using a small network directly to predict the offset field. After alignment, we can get refined rotated features and more accurate priors, which are useful for the final head's precise classification and regression. Fig. 2 also shows the framework of a RFL.

C. Gaussian Distribution Sampling Optimal Transport Assignment (GSOTA)

Advanced label assignment has important progress in object detection in recent years. Specifically, OTA [39] analyzes the label assignment from a global perspective and formulates the assigning procedure as an OT problem, and produces the SOTA performance that exceeds the current famous assigning strategies [24], [38], [40].

1) Gaussian Center Sampling Method: The original OTA method uses a strategy named Center Prior, those anchors which are not in the center distance are punished with a big value to the cost matrix. The SimOTA [20] label assignment method uses center sampling to replace the Center Prior strategy in OTA. This modification not only has similar effects as the Center Prior strategy in the label assignment problem but also can reduce the demander numbers in the OT problem, therefore the computational burden is also reduced.

However, the center sampling method in the SimOTA is designed for HBB, and it is ineffective when directly used with OBB, due to the following two reasons: First, the horizontal center sampling region is usually a 3×3 or 5×5 square. The OBB used to accurately surround the rotated target has a rotated angle compared with the traditional HBB. And square center sampling without rotation angle cannot fit OBB well. Second, there are a large number of targets with a large aspect ratio in aerial images, which leads to the sharp reduction of the short edge of the OBB, and the use of square center sampling does not reflect this feature. Intuitively, center sampling points should be in the ellipse center region with a certain range inside the oriented targets. This idea is shown in Fig. 3.

Hence, we propose a Gaussian center sampling (GCS) method based on 2-D Gaussian distribution modeling. First, we use Gaussian distribution to replace the OBB $\mathbf{OBB} = (cx, cy, w, h, \theta)$ as follows:

$$\Sigma = \mathbf{R}(\theta)\Sigma_{\mathbf{0}}\mathbf{R}^{\mathbf{T}}(\theta) \quad \mathbf{m} = (cx, cy) \tag{3}$$

$$\mathbf{R}(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \quad \Sigma_{\mathbf{0}} = \begin{pmatrix} \frac{w^2}{4} & 0 \\ 0 & \frac{h^2}{4} \end{pmatrix} \tag{4}$$

where $\mathbf{R}(\theta)$ is rotation matrix, and $\Sigma_{\mathbf{0}}$ represents the diagonal matrix of eigenvalues (i.e., the covariance matrix when angle equal to 0), $\mathbf{m}$ is mean value, and Σ is covariance matrix. So, the probability density of 2-D Gaussian distribution is given by

$$f(\mathbf{x}|\mathbf{m}, \Sigma) = \frac{1}{2\pi|\Sigma|^{\frac{1}{2}}}\exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^{\mathbf{T}}\Sigma^{-\mathbf{1}}(\mathbf{x}-\mathbf{m})\right) \tag{5}$$

where $\mathbf{x}$ indicates coordinates (x, y) in the image.The contour of the probability density function of 2-D Gaussian distribution is an elliptic curve, hence we can remove the normalization term from $f(\mathbf{x}|\mathbf{m}, \Sigma)$ get $g(\mathbf{x}|\mathbf{m}, \Sigma)$ as

$$g(\mathbf{x}|\mathbf{m}, \Sigma) = \exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m})^{\mathbf{T}}\Sigma^{-\mathbf{1}}(\mathbf{x}-\mathbf{m})\right) \tag{6}$$

when $(-(1/2)(\mathbf{x}-\mathbf{m})^{\mathbf{T}}\Sigma^{-\mathbf{1}}(\mathbf{x}-\mathbf{m})) = 1$, we can get a standard rotated ellipse equation, and this ellipse contour is the inscribed ellipse of the OBB.

Hence, the elliptic contour of 2-D Gaussian distribution in the OBB can be modeled as $g(\mathbf{x}|\mathbf{m}, \Sigma) \geq \exp(-0.5)$. That means the center sampling area of the oriented target can be determined by $g(\mathbf{x}|\mathbf{m}, \Sigma) \geq \exp(-0.5)$. For example, if $g(\mathbf{x})$ is greater than $\exp(-0.5)$, then the corresponding point $\mathbf{x}$ is in the OBB inner area and should be sampled.

When the target has a large aspect ratio, it makes the part in a long-axis direction far away from the center area. We should modify the Gaussian distribution to shrink the ellipse center sampling region. Therefore, the new Gaussian distribution can be derived by a modified covariance matrix as

$$\Sigma = \mathbf{R}(\theta)\left(\frac{\min(w,h)}{4}\begin{pmatrix} w & 0 \\ 0 & h \end{pmatrix}\right)\mathbf{R}^{\mathbf{T}}(\theta). \tag{7}$$

In this new Gaussian distribution, when $g(\mathbf{x}|\mathbf{m}, \Sigma) = \exp(-0.5)$ the length of ellipse major axis shrinks to $(wh)^{1/2}$ and minor axis remains unchanged. By using the long edge definition format of OBB, we can directly calculate the shrink covariance matrix as

$$\Sigma = \begin{pmatrix} \frac{wh}{4}\cos^2\theta + \frac{h^2}{4}\sin^2\theta & \cos\theta\sin\theta\left(\frac{wh}{4} - \frac{h^2}{4}\right) \\ \cos\theta\sin\theta\left(\frac{wh}{4} - \frac{h^2}{4}\right) & \frac{wh}{4}\sin^2\theta + \frac{h^2}{4}\cos^2\theta \end{pmatrix}. \tag{8}$$

Fig. 3 shows this ellipse center region of OBB. Compared with the horizontal center sampling, the GCS is more suitable for OBB, and using the shrink version for large aspect ratio target condition is more concentrated on truly center and gets more meaningful sampling points.

In addition, for decoupled detection heads, a certain level of spatial misalignment exists in predictions between the classification and regression tasks [42]. When some objects' geometric center has a large number of backgrounds or it is difficult to capture effective representation features in the central range, the spatial misalignment between these two tasks is obvious. We propose a very simple solution, called the maximum classification center mean (MCCM), which does not need external positive sample reweight adjustment (like [43]'s solution), and can be directly integrated into our GCS method.

The idea of the MCCM is as follows: First, the point in ground-truth bounding boxes is taken as the candidate

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sampling point. Then, according to the prediction results of classification branches, the maximum classification prediction probability at each candidate sampling point is taken as the probability value, and the center coordinate of the selected points is selected as the new mean value in the Gaussian distribution, the selected point corresponds to the maximum probability value in the same ground-truth bounding boxes. At last, get the new 2-D Gaussian distribution, and the discrimination function g is updated. The points within this new Gaussian distribution and still in ground truth are selected as sampling points. The pseudo-code of our proposed GCS with MCCM is shown in Algorithm 1.

Algorithm 1 GCS

Input:
P_I is the set of total candidate sampling points
G^{box} is the gt OBB annotations for objects, represent as (cx, cy, w, h, θ)
S^{cls} is the network predicted multiclassification scores for each candidate sampling points
Output:
P_O is the set of total actually sampling points
$\Sigma \leftarrow$ calculate shrink covariance matrix base on G^{box}'s (w, h, θ)
$\mathbf{m} \leftarrow (cx, cy)$
$P_O \leftarrow [\ ]$
for p **in** P_I **do**:
  if $g(p|\mathbf{m}, \Sigma)$(Eq.6) $\geq \exp(-0.5)$:
    $P_O \leftarrow$ append(p)
  if p in G^{box}:
    $P' \leftarrow$ append(p)
for s **in** S^{cls} **do**:
  $s \leftarrow$ Maximum classification probability in s
  for gt **in** G^{box} **do**:
    $p' \leftarrow P'$ in gt
    $s' \leftarrow s$ corresponding to p'
    max_indexes $\leftarrow$ argmax(s')
    $(cx', cy') \leftarrow p'$[max_indexes]
    $\mathbf{m} \leftarrow (cx', cy')$
for p **in** P' **do**:
  if $g(p|\mathbf{m}, \Sigma)$(Eq.6) $\geq \exp(-0.5)$:
    $P_O \leftarrow$ append(p)
return P_O

2) Use Improved Dynamic Top-k Strategy: In practice, via the Sinkhorn-Knopp algorithm to solve the OT problem like the original OTA method is quite expensive for model training. Therefore, we follow the SimOTA method, which uses a dynamic top-k strategy to get an approximate solution for the OT problem.

In our experiments, we found original dynamic top-k strategy has a convergent slow problem when directly used in aerial image detection tasks, as shown in Fig. 4. We think this is because in the init few epochs, the IoU values of the model predict OBBs and ground-truth bounding boxes are too small. Specifically, the dynamic k estimation method is used in the dynamic top-k strategy to obtain the number of positive

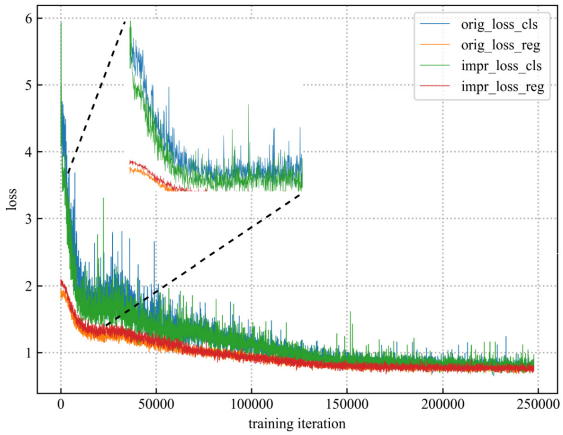

Fig. 4. Comparison curve of training loss for using impr-DkE and using the original method. The blue and orange curves represent the classification branch loss and regression branch loss, respectively, by using the original method. The green and red curves represent the classification branch loss and regression branch loss, respectively, by using the original method.

samples to be sampled, and this method is dependents on the IoU value to calculate the number of positive samples k. Therefore, in the first few epochs, as shown in Fig. 7(a), most ground truth is assigned one positive sample point, which means the model is equivalent to executing the assignment strategy of one-to-one, which greatly reduces the efficiency of the optimization algorithm in updating the model parameters, and thus causes the problem of slow convergence. Hence, this article proposes an improved dynamic k estimation (impr-DkE) method, to be used in the initially designated epoch of model training. Specifically, the candidate IoU value is squared on our impr-DkE method, so the IoU value is amplified, which enables more positive samples to be obtained. After exceeding the designated epoch, the original IoU value is used. This improvement method's pseudo-code is shown in Algorithm 2.

Algorithm 2 Improved Dynamic k Estimation

Input:
c_k is the number of candidates maximum IoU
iou are the pairwise IoU values of GCS sampled oriented bounding boxes and the gound-truth box
$init$ is a bool signal that this method is used in the first few init epochs
Output:
k is the number of gound-truth's dynamic positive samples.
$nums \leftarrow \min(c_k, \text{len}(iou))$
$topk_iou \leftarrow \text{topk}(iou, nums)$
if $init$:
  $k \leftarrow \text{clamp}(\text{int}(\text{sum}(sqrt(topk_iou))), min = 1)$
else:
  $k \leftarrow \text{clamp}(\text{int}(\text{sum}(topk_iou)), min = 1)$
return k

3) Proposed Assignment Method: Combining the GCS with MCCM method and the improved dynamic top-k strategy, we propose the GSOTA method. For detail, GSOTA first uses the GCS method to extract sample points that are in the center

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region of targets, which will massively reduce the demander numbers on the OT problem, in other words, the GCS method simplifies the OT problem. Second, in the context of object detection, supposing there are m ground-truth targets and n points from GCS extracted sampling, we view each gt as a supplier which holds k units of positive labels, and each point as a demander which needs one unit of labels. Calculate the pairwise matching degree of the candidate sampling points extracted by GCS and ground-truth targets, represented by the cost matrix. For example, the cost between ground-truth gt_i and sampling point p_j is calculated as

$$c_{ij} = L_{ij}^{\text{cls}} + \lambda L_{ij}^{\text{reg}} \tag{9}$$

where λ is a balancing coefficient, in our experiments, it is set to 3. L_{ij}^{cls} and L_{ij}^{reg} are classification loss and regression loss between gt_i and sampling point p_j. Then, for each gt_i, we use the improved dynamic top-k strategy to select dynamic k sampling points with the least cost as its positive samples. Repeat this process until all gt is assigned, the remaining points that are not gt assigned are treated as negative samples, and the other points not selected by GCS are also assigned as negative samples.

As shown in Table I, our proposed GSOTA compared with SimOTA raises the detector performance from 66.32 to 69.36 mAP, showing the effectiveness of our assigning method in the rotated object detection problem. The GSOTA pseudo-code is shown in Algorithm 3.

Algorithm 3 GSOTA

Input:
I is an input image
P is the set of total candidate sampling points
G is the gt annotations for objects in image I
λ is the balanced coefficient in Eq.9
Output:
π is the optimal assigning plan
$m \leftarrow |G|$
$P \leftarrow GCS(P, G^{box})$
$n \leftarrow |P|$
$Pre^{cls}, Pre^{box} \leftarrow$ Network Forward(I)
pairwise cls cost : $c_{i,j}^{cls} = L^{cls}(Pre_j^{cls}, G_i^{cls})$
pairwise reg cost : $c_{i,j}^{reg} = \lambda L^{reg}(Pre_j^{box}, G_i^{box})$
$cost = c^{cls} + c^{reg}$
$\pi_{i,j}(i = 1, 2, \ldots, m;\ j = 1, 2, \ldots, n) \leftarrow$ ZerosInit
$k_i(i = 1, 2, \ldots, m) \leftarrow$ imporved Dynamic k Estimation
for $i = 1$ **to** m **do**:
  $index_set \leftarrow \text{topk}(cost_{i,:}, k_i, \text{largest} = \text{False})$
  $\pi_{i,index_set} \leftarrow 1$
for $j = 1$ **to** n **do**:
  if $\text{sum}(\pi_{:,j}) > 1$:
    $\pi_{:,j} \leftarrow 0$
    $index \leftarrow \min(cost_{:,j})$
    $\pi_{index,j} \leftarrow 1$
return π

D. Proposed Detector (R2YOLOX)

We adopt rotated YOLOX as the baseline, including its network architecture and most parameter settings, our R2YOLOX is based on the combination of RRM and GSOTA, and apply these methods on the baseline. R2YOLOX's whole structure is shown in Fig. 1. In the following, we detail R2YOLOX in both the training and inference phase.

1) Regression Targets: The RRM predicts five parameters, i.e., four distances from each point on the feature map to the left, top, right, and bottom sides of ground-truth boxes, and the rotated angle of the ground-truth box. Because the ground-truth coordinate system is unparallel to the image coordinate system, we need to transform each point on the feature map into its corresponding ground-truth coordinate system. Hence, the RRM parameterized regression target as

$$(x_r, y_r) = R^{-1}(\theta_g)\left(x - x_g, y - y_g\right)^{\text{T}} \tag{10}$$

$$(t_l, t_t, t_r, t_b) = \log\left(\frac{\frac{w_g}{2} + x_r}{\text{stride}_i}, \frac{\frac{h_g}{2} + y_r}{\text{stride}_i}, \frac{\frac{w_g}{2} - x_r}{\text{stride}_i}, \frac{\frac{h_g}{2} - x_r}{\text{stride}_i}\right) \tag{11}$$

$$t_\theta = \theta_g / \pi \tag{12}$$

$$(w^*, h^*) = \left(\left(e^{t_l^*} + e^{t_r^*}\right), \left(e^{t_t^*} + e^{t_b*}\right)\right)\text{stride}_i \tag{13}$$

$$(x^*, y^*) = \frac{\text{stride}_i}{2}\left(\left(e^{t_l^*} - e^{t_r^*}\right), \left(e^{t_t^*} - e^{t_b^*}\right)\right) R(\theta^*) \tag{14}$$

$$\theta^* = \pi t_\theta^* \tag{15}$$

where x, y denote the center coordinates of the feature map grid point, stride_i is the feature map grid stride and i corresponds to the level of feature pyramid network (FPN), and the rotated matrix is the same as (4). Variables x_g, y_g, h_g, w_g, and θ_g denote the ground-truth box's center coordinates, width, height, and rotated angle, respectively. $t_l^*, t_t^*, t_r^*, t_b^*$, and t_θ^* are parameters directly predicted by the RRM network. x^*, y^*, w^*, h^*, and θ^* are the decoded prediction box.

The regression outputs of RRM (the RRM decoded prediction bounding box) as the refined priors are used to align rotated features and final prediction decode. Hence, following previous works, the final decoupled head's regression targets as

$$\left(t_x, t_y\right) = \left(x_g - x, y_g - y\right) R(\theta)\left(\frac{1}{w}, \frac{1}{h}\right) \tag{16}$$

$$(t_w, t_h) = \left(\log\left(\frac{w_g}{w}\right), \log\left(\frac{h_g}{h}\right)\right) \tag{17}$$

$$t_\theta = \left(\theta_g - \theta\right)/\pi \tag{18}$$

$$\left(x^*, y^*\right) = \left(t_x^* w + x, t_y^* h + y\right) R^{-1}(\theta) \tag{19}$$

$$\left(w^*, h^*\right) = \left(we^{t_w^*}, he^{t_h^*}\right) \tag{20}$$

$$\theta^* = \pi t_\theta^* + \theta \tag{21}$$

where x, y, w, h, θ, and $R(\theta)$ are the refined priors' center coordinates, width, height, rotated angle, and rotation matrix, respectively. The rotated matrix is the same as (4). And x_g, x^* are for ground-truth box and predicted box (likewise for y, w, h, θ).

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2) Loss Function: The loss of R2YOLOX consists of two parts, i.e., the loss of RRM and the loss of the final decoupled head. For each part, we use the GSOTA method to assign a class label to each point/refined prior and regress its location. The loss function can be defined as

$$\mathcal{L} = \frac{1}{N_R}\left(\lambda_1 \sum_i \mathcal{L}_{\text{cls}}^R\left(p_i^R, l_i\right) + \lambda_2 \sum_i \mathcal{L}_{\text{reg}}\left(b_i^R, g_i\right)\right) + \frac{1}{N}\left(\lambda_3 \sum_i \mathcal{L}_{\text{cls}}\left(p_i, l_i\right) + \lambda_4 \sum_i \mathcal{L}_{\text{reg}}\left(b_i, g_i\right)\right) \quad (22)$$

where N_R, N indicate the number of positive samples in the RRM and the final decoupled head, respectively, i is the index of a positive sample in a minibatch, p_i^R and b_i^R are the predicted category and decode bounding box of the point i in RRM, and p_i and b_i are the predicted object category and decode bounding box in the final decoupled head. l_i represents the ground-truth category of ith object and g_i is the ground-truth bounding box. $\lambda_1, \lambda_2, \lambda_3$, and λ_4 are loss balance parameters, which are set to {5, 5, 2, 5} by default. The CrossEntry loss is adopted as the classification loss $\mathcal{L}_{\text{cls}}$ in the final head as same as the YOLOX. For RRM, we use quality focal loss (QFL) [44] as the $\mathcal{L}_{\text{cls}}^R$, which can effectively use the IoU information to suppress low-quality detection results and reduce the spatial misalignment of classification and regression. Due to the ordinary IoU loss cannot be directly used in rotated object bounding box regression, we use KLD loss to replace the ordinary IoU loss as the regression loss $\mathcal{L}_{\text{reg}}$, which is a kind of IoU loss but specially designed for rotated object detection tasks.

3) Inference: In the model inference, R2YOLOX can simply forward an image through the whole network without any complex RoI operation and complex label assignment method. We just choose the top-2000 predictions based on their classification scores. The threshold of rotated NMS is set to 0.1 to yield the final detection results.

IV. Experiments

Our method was evaluated on four challenging aerial object detection datasets, i.e., DOTA, HRSC2016, UCAS-AOD, and FGSD2021.

A. Datasets

DOTA [1] is a large-scale dataset for aerial object detection, which contains 2806 large aerial images with a size range from 800 × 800 to 4000 × 4000 and 1 88 282 instances of 15 common object categories including Plane (PL), Baseball diamond (BD), Bridge (BR), Ground track field (GTF), Small vehicle (SV), Large vehicle (LV), SH, Tennis court (TC), Basketball court (BC), Storage tank (ST), Soccer-ball field (SBF), Roundabout (RA), Harbor (HA), Swimming pool (SP), and Helicopter (HC). Both train and validation sets are used for training and the test set is used for testing. Following [45], we crop a series of 1024 × 1024 patches from original images with an overlap of 200. For multiscale experiments, we resize original images at three scales (0.5, 1.0, and 1.5) and then crop them into 1024 × 1024 patches with an overlap of 512.

HRSC2016 [4] is an SH recognition dataset annotated with OBB, which contains 1061 aerial images, and the image size ranges from 300 × 300 to 1500 × 900. We use both the training (436 images) and validation (181 images) sets for training and the testing set (444 images) for testing. All images are resized to 800 × 800 without changing the aspect ratio.

UCAS-AOD [46] is an aerial image dataset for oriented small object detection, including care and PL. It contains 1510 images with 510 car images and 1000 PL images and 14 596 instances in total. In line with Oriented RepPoints [18] and the other methods, we randomly divide it into the training set, the validation set, and the test set with a ratio of 5:2:3. The size of all images is resized to 800 × 800.

FGSD2021 [34] is a fine-grained SH detection dataset, which annotates many different types of SHs with OBB. FGSD2021 total collected 636 images with a fixed GSD (1 m per pixel), including 5274 labeled targets and 20 categories. We follow CHPDet [34]'s setting divide this dataset into 424 training images and 212 test images and cropped all training set images and testing set images into 1024 × 1024 slices with a stride of 824, resized them to 800 × 800, like our previous processing of datasets.

B. Implement Details

We adopt CSPDarknet with cross stage partial (CSP)-path aggregation feature pyramid network (PAFPN) as the backbone of R2YOLOX. In detail, we use CSPDarknet-L and CSPDarknet-X as backbones for fair comparisons with other methods and CSPDarknet-S for other experiments if not specified. For most experiments, we follow the original YOLOX series models training config, adopting random flipping and rotation, Mosaic, and MixUp augmentation to avoid overfitting, and these augmentations are also the key to training a high-performance lightweight model. For the testing phase, we do not use any augmentations. The hyperparameters of KLD loss $\mathcal{L}_{\text{reg}}$ are set as default, i.e., $\tau = 1$, and use log nonlinear function in our models. All experiments are conducted on an NVIDIA RTX 3080Ti with a batch size of 2. Because of the limitation of our computing resources, we have to resize all input images from 1024 × 1024 to 800 × 800 on the DOTA dataset. In the training phase, the stochastic gradient descent (SGD) optimizer is adopted with an initial learning rate of 1.25×10^{-3}, with the momentum of 0.9 and the weight decay of 1.0×10^{-4}. We train all models in 40 epochs for DOTA, HRSC2016, UCAS-AOD, and FGSD2021 and adopt the same training schedules as MMDetection [47]. Our model is based on MMDetection and open MultiMedia Lab rotate detection toolbox and benchmark (MMRotate) open source libraries, which are two object detection toolboxes based on the PyTorch framework.

C. Ablation Studies

In this section, we perform a series of experiments on the DOTA to validate the effectiveness of our proposed method.

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TABLE I

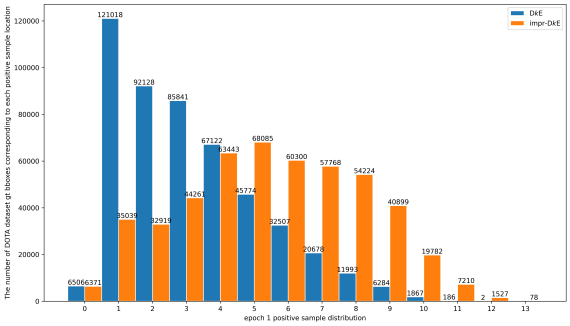
RESULT OF ABLATION EXPERIMENTS FOR R2YOLOX-S ON DOTA WITH SINGLE-SCALE TRAINING AND TESTING. √ MEANS THE MODULE IS USED. WE CHOOSE THE ROTATED YOLOX-S AS THE BASELINE, RRM IS REFINED ROTATED MODULE, GSOTA MEANS GAUSSIAN DISTRIBUTION SAMPLING OPTIMAL TRANSPORT ASSIGNMENT

Methods	RRM	GSOTA	mAP (%)
Baseline			66.32
R2YOLOX-S	√		68.19
		√	69.36
	√	√	72.25

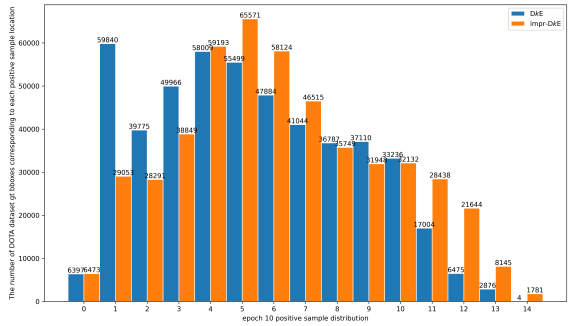
TABLE II

RESULT OF ABLATION EXPERIMENTS FOR R2YOLOX-S ON DOTA WITH SINGLE-SCALE TRAINING AND TESTING. √ MEANS THE MODULE IS USED. WE CHOOSE THE ROTATED YOLOX-S AS THE BASELINE, RRM IS REFINED ROTATED MODULE. 2 × RRM MEANS CASCADE RRM TWICE

Methods	RRM	2×RRM	mAP (%)
Baseline			66.32
R2YOLOX-S	√		68.12
		√	68.19



(a)



(b)

Fig. 5. Comparison histogram of the distribution of GT assigned positive sample number of original method and impr-DkE. (a) Shows comparison distributions at epoch 1 training. (b) Epoch 10 training.

CSPDarknet-S with CSP-PAFPN is adopted as the backbone and calls this model R2YOLOX-S. Other models in the R2YOLOX series can be found in Section IV-D. We choose the rotated YOLOX model as our baseline, which achieves 66.32% mAP on DOTA. The detail of the rotated YOLOX can be found in Section III-A. All ablation experiments are run on single-scale training and testing.

1) Evaluation on RRM: Because the one-stage detector has inherent feature misalignment, there is a huge gap in the performance between using the YOLOX model on the oriented object detection task and on the general object detection task. Therefore, we introduce RRM and use AlignConv to alleviate the problem of feature misalignment by referring to the architecture in S^2ANet [11]. After adding RRM to the baseline, the performance gains 1.8% to 68.12% mAP, as shown in Table II. However, adding RRM makes the structure of the model complex; moreover, we found that using cascade RRM architecture which is similar to casade-S^2ANet does not get any significant performance improvement, and the result is in Table II. We believe that this phenomenon may be due to the fact that the problem of feature misalignment is solved well by using one layer of AlignConv, and the second layer of AlignConv on the cascade RRM structure cannot play an obvious role. In addition, stacking multiple RRMs causes the number of model learning parameters to increase, making the model difficult to converge and needing more longer training epochs. Hence, we only add one RRM to the model in the rest of the experiments.

2) Evaluation on GSOTA: Choosing appropriate positive point set samples plays a key role in training the high-performance one-stage anchor-free aerial object detector. Therefore, we proposed an advanced oriented label assignment method GSOTA, and the original OTA assignment method is improved from following two aspects: limiting the sampling region and accelerating the model training. Our GSOTA achieves significant improvement in rotated detection performance.

For getting the appropriate sampling region, we proposed the GCS method using the Gaussian distribution center sampling region to replace the fixed rectangular center sampling region, this simple idea is shown in Fig. 3. However, in the model training process, we found that there exists a certain degree of spatial misalignment in classification and regression tasks, and MCCM method is proposed to solve this spatial misalignment problem by enlarging the sampling range without external reweight parameters. Table III shows that the mAP could increase to 67.99% by using GCS, and by using GCS with MCCM the performance could increase to 68.86%. Figs. 6 and 7 show sampling region results of our proposed GCS and GCS with MCCM, at the start of training and the end of training. From these two Figs. 6 and 7, we can see that GCS with MCCM can cover almost all the ranges in the feature map that are conducive to model learning. And as the training progress, this sampling region will be more and more accurate, which makes the model easy to get appropriate positive sample points and conducive to training convergence.

From the perspective of accelerating model training, we proposed impr-DkE to get more positive sample points compared with the original dynamic k estimation, as shown in Fig. 5. Compared with the original method, at epoch 1 [Fig. 7(a)], by using the impr-DkE method, the number of ground truths allocated to five positive samples is the largest, instead of being assigned to only one positive sample. And the distribution of the number of positive samples obeys the normal distribution approximately, which may more conducive to the supervised

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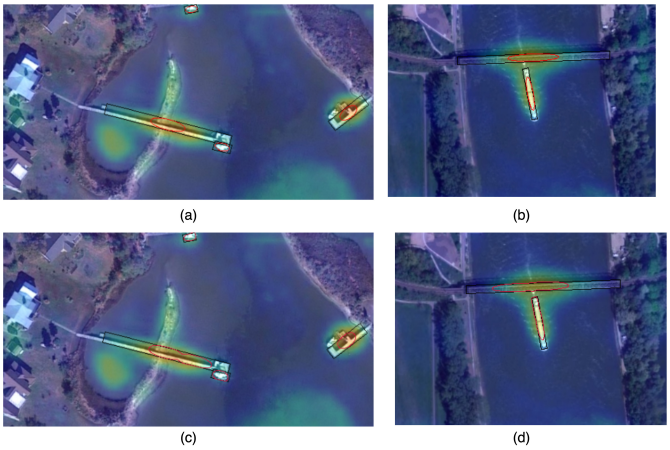

Fig. 6. Examples of GCS sampling region at epoch 1. The black rectangular box indicates ground-truth bounding boxes. The red ellipse region in (a) and (b) subfigures indicates the GCS sampling region. The red irregular region in (c) and (d) indicates the GCS with the MCCM sampling region. Each subfigure is generated by superimposing the Heatmap output from the FPN level 0 final classification branch of the model and the original image.

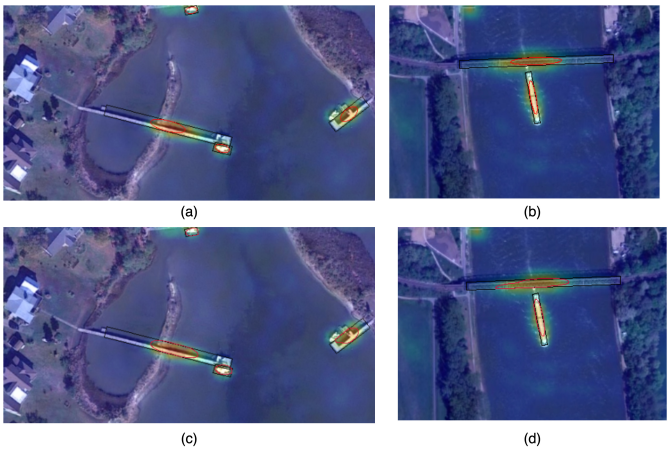

Fig. 7. Examples of GCS sampling region at epoch 40. The black rectangular box indicates ground-truth bounding boxes. The red ellipse region in (a) and (b) subfigures indicates the GCS sampling region. The red irregular region in (c) and (d) indicates the GCS with the MCCM sampling region. Each subfigure is generated by superimposing the Heatmap output from the FPN level 0 final classification branch of the model and the original image.

learning of the model. At epoch 10 [Fig. 7(b)], we find that the distribution of positive samples allocated by the original method is more and more similar to that using the impr-DkE method. We think this is because, with the training of the model, the IoU value between predicted rotated bounding boxes by the model and ground-truth bounding boxes becomes larger and larger; therefore, the original method can also obtain a sufficient number of positive samples for the model learning. So, in the later training epoch, we switch back to using the original dynamic k estimation method.

Fig. 4 shows that our proposed impr-DkE gets a small classification loss and the classification loss curve is always ahead of the original loss curve, which means using impr-DkE could early converge and accelerate training. In addition, as shown in Table III, by combining GCS and impr-DkE we can achieve 69.36% mAP. Hence, our proposed method is not only beneficial to accelerating the convergence of model training but also improves detection performance. For

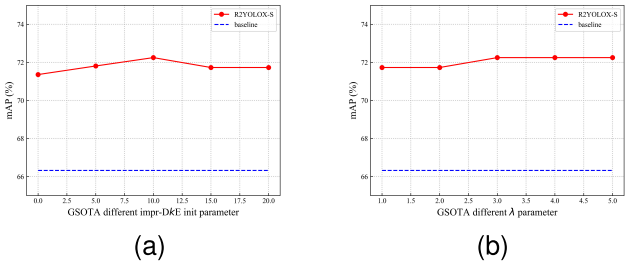

Fig. 8. Evaluation of hyperparameters. (a) Shows experimental results under different values of the hyperparameter init on the impr-DkE method. (b) Shows experimental results under different values of the hyperparameter λ on GSOTA.

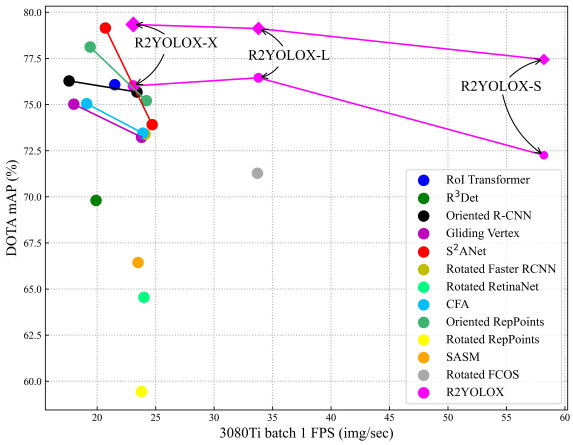

Fig. 9. Speed of all methods is reported on one 3080Ti graphics processing unit (GPU) in terms of FPS. It is noted that Gliding Vertex, Oriented R-CNN, S^2ANet, convex-hull feature adaptation (CFA), and Oriented RepPoints are using ResNet-50 and ResNet-101, and the other methods just using ResNet-50. In addition, the S^2ANet and our R2YOLOX also show multiscale training and testing results, all methods are reimplemented by the MMRotate framework.

TABLE III

RESULT OF ABLATION EXPERIMENTS FOR R2YOLOX-S ON DOTA WITH SINGLE-SCALE TRAINING AND TESTING. √ MEANS THE MODULE IS USED. WE CHOOSE THE ROTATED YOLOX-S AS THE BASELINE. GCS MEANS GAUSSIAN CENTER SAMPLING METHOD. W/MCCM INDICATES GCS COMBINED WITH MCCM, AND THE IMPR-DkE MEANS IMPROVED DYNAMIC k ESTIMATION

Methods	GCS	w/MCCM	impr-DkE	mAP (%)
Baseline				66.32
R2YOLOX-S	√			67.99
	√	√		68.86
	√	√	√	69.36

convenience, in the following article, we use the GSOTA to represent our improved method of OTA, which combines GCS with MCCM and impr-DkE.

As shown in Table I, our proposed improved methods RRM and GSOTA are independent of each other, combining RRM and GSOTA, the mAP can achieve 72.25% mAP. These results demonstrate that our proposed methods are useful.

Because there are two hyperparameters init, and λ, in GSOTA, hence, we conducted experiments to find the appropriate values of these two hyperparameters of GSOTA when combining with RRM, as shown in Fig. 8. We experimentally validated that the best value of parameter init is ten epochs on the DOTA datasets [in Fig. 9(a)], this is also confirmed by the

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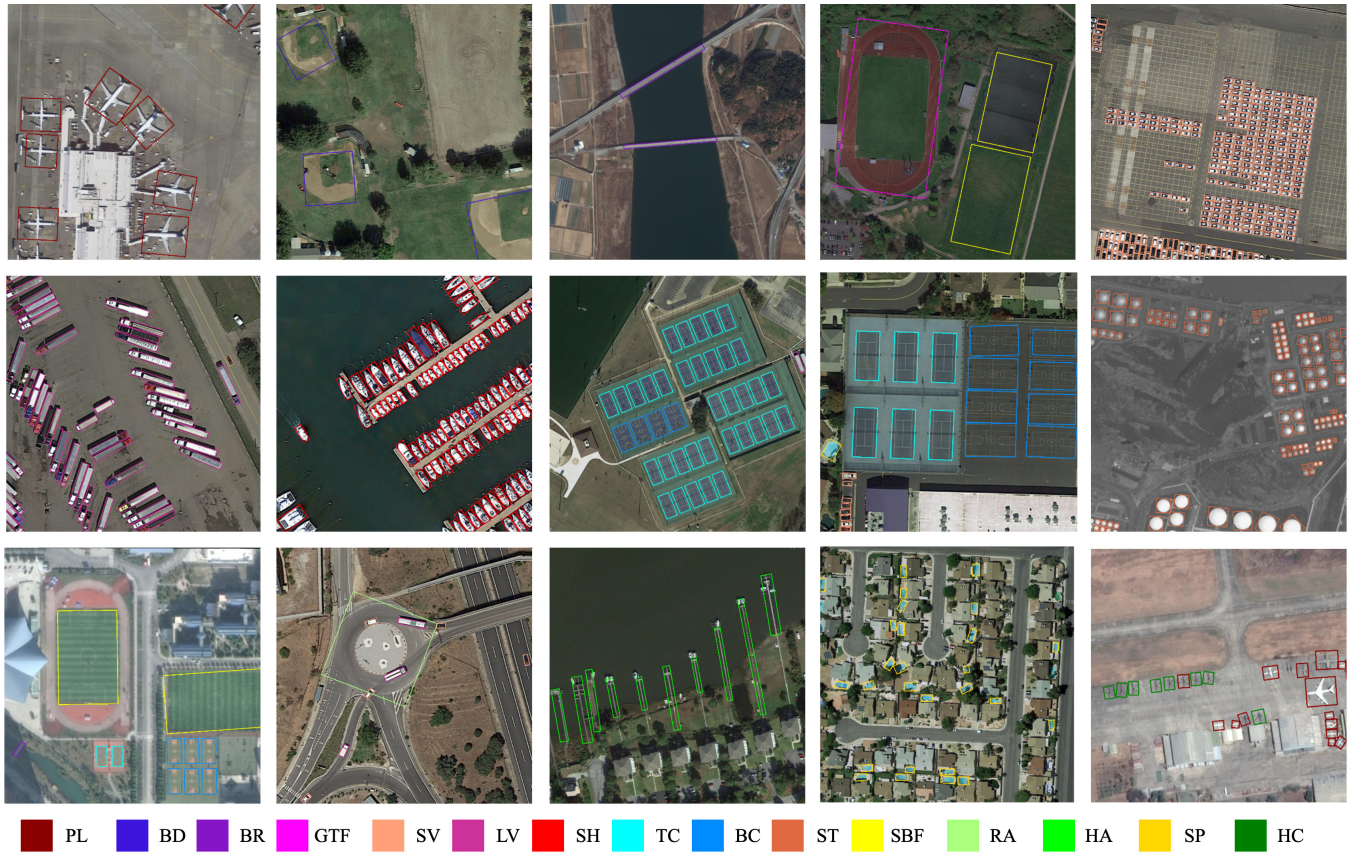

Fig. 10. Some detection results of our proposed R2YOLOX-X by the single-scale training and testing on the DOTA dataset. The confidence threshold is set to 0.3 when visualizing these results, and one color stands for one object class.

analysis of the positive sample distribution in Fig. 5. And the best value of λ is 3 like the original OTA method. We also found the different λ has little effect on the performance of the model [in Fig. 9(b)], we think this is because the cost matrix on the GSOTA aggregates the regression and classification branches' cost through addition instead of multiplication [42]; therefore, the λ has no significant impact on the final positive sample assignment results.

D. More Backbones

Following the YOLOX scaling rule, we modified the CSPDarknet-S to produce CSPDarknet-L and CSPDarknet-X, we use these backbones to replace CSPDarknet-S reconstruct the R2YOLOX series model, named R2YOLOX-L and R2YOLOX-X. The model parameters, floating-point operations Per second (FLOPs), FPS, and mAP, of our proposed model on DOTA are shown in Table IV. The results in Table IV show that our proposed models not only ensure high accuracy but also achieve the SOTA inference speed of ODAIs. Our R2YOLOX-S model has the smallest parameter quantity only 11.16 M parameters and 28.58 giga floating-point operations per second (GFLOPs), as far as we know, it reaches the highest 58.2 FPS on our device, just using FP32. These are important factors to the actual deployment and the use of the deep learning model on the UAV platform. The comparison of our proposed models' speed-accuracy trade-off with other models is shown in Fig. 9. Our model achieves the fastest speed and competitive accuracy, which means, our model has the best speed-accuracy trade-off on the DOTA.

TABLE IV

R2YOLOX SERIES MODEL PARAMETER, GFLOPS, FPS, AND MAP COMPARISON. THE MAP IS THE RESULT OF DOTA WITH SINGLE-SCALE TRAINING AND TESTING. FPS AND FLOPS ARE RESULTS OF INFERENCE WITH 1024 × 1024 IMAGE SIZE WITH A SINGLE 3080TI CONDITION, AND JUST USING FP32

Methods	Backbone	Param	GFLOPs	FPS	mAP (%)
R2YOLOX-S	CSPDarknet-S	11.16M	28.58	58.2	72.25
R2YOLOX-L	CSPDarknet-L	63.01M	152.37	33.8	76.45
R2YOLOX-X	CSPDarknet-X	112.86M	268.51	23.1	76.01

E. Comparison With SOTAs

In this section, we compare our proposed R2YOLOX with other SOTA methods on four challenging aerial detection datasets, i.e., DOTA, HRSC2016, UCAS-AOD, and FGSD2021.

1) Results on DOTA: As shown in Table V, we compare our R2YOLOX series method with other SOTA methods on DOTA task 1 (i.e., OBB detection task). We achieve the best 79.33% mAP through the R2YOLOX-X model, this is a SOTA performance of anchor-free methods on the DOTA, which can compete with many famous anchor-based two-stage and one-stage methods. Compared with other methods, our R2YOLOX model produces better detecting accuracy on the HC, BR, Habor, SH, SP, TC, BC, and soccer-ball-field detection tasks. Some qualitative detection results of our proposed R2YOLOX

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TABLE V

COMPARISONS WITH SOTA RESULTS ON DOTA. R-101-FPN STANDS FOR RESNET 101 WITH FPN (LIKEWISE R-50-FPN), H-104 STANDS FOR HOURGLASS 104, AND DEEP LAYER AGGREGATION (DLA)-34 REPRESENTS THE DLA NETWORK. AND ALL RESULTS ARE REPORTED ON SINGLE-SCALE TRAINING AND TESTING, IF NOT SPECIFICALLY. † MEANS USE MULTISCALE TRAINING AND TESTING. THE BEST AND SECOND-BEST RESULTS IN EACH CATEGORY ARE HIGHLIGHTED IN BOLD

Methods	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP (%)
Anchor-base two stage																	
ICN [48]	R-101-FPN	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
Faster R-CNN-O [45]	R-50-FPN	88.44	73.06	44.86	59.09	73.25	71.49	77.10	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05
RoI Transformer [7]	R-101-FPN	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CenterMap-Net [9]	R-50-FPN	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	72.36	58.70	71.74
SCRDet [13]	R-101-FPN	**89.98**	80.65	52.09	68.36	68.36	60.32	72.41	90.85	87.94	86.86	65.02	66.68	66.25	68.24	65.21	72.61
Gliding Vertex [10]	R-101-FPN	89.64	85.00	52.26	**77.34**	73.01	73.14	86.82	90.74	79.02	86.81	59.55	**70.91**	72.94	70.86	57.32	75.02
ReDet [49]	ReR50	88.79	82.64	53.97	74.00	78.13	84.06	88.04	90.89	87.78	85.75	61.76	60.39	75.96	68.07	63.59	76.25
Oriented R-CNN [12]	R-101-FPN	88.86	83.48	55.27	76.92	74.27	82.10	87.52	**90.90**	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
Anchor-base one stage																	
RetinaNet-O [45]	R-50-FPN	88.67	77.62	41.81	58.17	74.58	71.64	79.11	90.29	82.18	74.32	54.75	60.60	62.57	69.67	60.64	68.43
DAL [28]	R-50-FPN	88.68	76.55	45.08	66.80	67.00	76.76	79.74	90.84	79.54	78.45	57.71	62.27	69.05	73.14	60.11	71.44
R^3Det [27]	R-101-FPN	88.76	83.09	50.91	67.27	76.23	80.23	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
S^2ANet [11]	R-50-FPN	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
S^2ANet† [11]	R-50-FPN	88.89	83.60	**57.74**	**81.95**	79.94	83.19	**89.11**	90.78	84.87	**87.81**	**70.30**	68.25	**78.30**	77.01	69.58	**79.42**
Anchor-free																	
PIoU [37]	DLA-34	80.90	69.70	24.10	60.20	38.30	64.40	64.80	**90.90**	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
DRN [16]	H-104	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
CFA [17]	R-101-FPN	89.26	81.72	51.81	67.17	79.99	78.25	84.46	90.77	83.40	85.54	54.86	67.75	73.04	70.24	64.96	75.05
PolarDet [50]	R-50-FPN	**89.73**	**87.05**	45.30	63.32	78.44	76.65	87.13	90.79	80.58	85.89	60.97	67.94	68.20	74.63	68.67	75.02
PolarDet† [50]	R-101-FPN	89.65	**87.07**	48.14	70.97	78.53	**89.34**	87.45	90.76	85.63	86.87	61.64	**70.32**	71.92	73.09	67.15	76.64
Oriented RepPoints [18]	R-50-FPN	86.95	80.84	53.20	70.50	80.85	79.35	87.36	90.88	85.95	85.60	61.12	70.08	73.43	72.58	56.46	75.68
Oriented RepPoints† [18]	R-101-FPN	88.72	80.56	55.69	75.07	**81.84**	82.40	87.97	90.80	84.33	**87.64**	62.80	67.91	77.69	**82.94**	65.46	78.12
Rotated-YOLOX (ours)	CSPDarknet-S	88.78	65.17	45.24	54.97	72.97	72.71	79.60	90.87	76.58	82.23	41.36	57.88	56.03	65.63	44.88	66.32
R2YOLOX-S (ours)	CSPDarknet-S	88.18	81.66	42.31	61.69	80.31	78.65	88.22	90.88	80.68	86.22	48.21	58.59	73.53	72.51	52.00	72.25
R2YOLOX-S (ours)†	CSPDarknet-S	87.10	84.12	50.96	70.82	80.92	**85.30**	88.51	90.88	**88.86**	87.54	60.94	63.03	77.92	71.45	73.29	77.44
R2YOLOX-L (ours)	CSPDarknet-L	89.13	84.42	48.63	68.46	81.61	79.34	88.45	**90.90**	87.18	87.00	57.59	64.75	77.01	**79.83**	62.43	76.45
R2YOLOX-L (ours)†	CSPDarknet-L	87.66	86.72	**57.74**	71.20	81.25	85.06	88.32	90.87	87.76	87.41	65.25	64.19	78.27	79.40	**75.71**	79.12
R2YOLOX-X (ours)	CSPDarknet-X	89.14	85.30	51.26	71.58	**82.10**	79.47	**88.58**	90.88	**88.56**	87.27	52.92	59.75	78.27	72.80	62.33	76.01
R2YOLOX-X (ours)†	CSPDarknet-X	88.12	85.93	55.38	76.61	80.46	85.21	88.45	90.88	88.51	87.44	**67.92**	61.13	**78.62**	77.69	**77.72**	**79.33**

TABLE VI

COMPARISON OF SOTA METHODS ON HRSC2016. THE BEST MAP RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Backbone	mAP(07)	mAP(12)
PIoU [37]	DLA-34	89.2	-
DRN [16]	H-104	-	92.7
SKNet [32]	H-104	90.4	-
SKFat [33]	H-104	**91.4**	-
R^3Det [27]	R-101-FPN	89.2	96.0
DAL [28]	R-101-FPN	89.7	-
S^2ANet [11]	R-101-FPN	90.1	95.0
Rotated RPN [15]	R-101-FPN	79.0	85.6
R^2CNN [51]	R-101-FPN	73.0	79.7
RoI Transformer [7]	R-101-FPN	86.2	-
Gliding Vertex [10]	R-101-FPN	88.2	-
CenterMap-Net [9]	R-50-FPN	-	92.8
PolarDet [50]	R-50-FPN	90.1	-
Oriented RepPoints [18]	R-50-FPN	90.3	97.2
R2YOLOX-S(ours)	CSPDarknet-S	89.8	96.0
R2YOLOX-L(ours)	CSPDarknet-L	89.8	96.8
R2YOLOX-X(ours)	CSPDarknet-X	90.1	**97.4**

TABLE VII

COMPARISON OF SOTA METHODS ON UCAS-AOD. THE BEST MAP RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Car	Plane	mAP(07)	mAP(12)
YOLOv3 [23]	74.6	89.5	82.1	-
RetinaNet [22]	84.6	90.5	87.5	-
	93.6	97.3	-	95.4
Faster R-CNN [14]	86.8	89.8	88.3	-
RoI Transformer [7]	88.0	90.0	89.0	-
RIDet-Q [52]	88.5	89.9	89.2	-
RIDet-O [52]	88.8	90.3	89.6	-
DAL [28]	89.2	90.5	89.8	-
S^2ANet [11]	89.5	90.4	89.9	-
DRBox [53]	85.0	94.9	-	89.9
S^2ARN [54]	92.2	97.6	-	94.9
ICN [48]	-	-	-	95.6
FADet [55]	92.7	98.7	-	95.7
R^3Det [27]	94.1	98.2	-	96.1
SCRDet [13]	94.9	98.9	-	96.9
PolarDet [50]	94.9	99.0	-	97.0
Oriented RepPoints [18]	89.5	90.7	**90.1**	-
R2YOLOX-S(ours)	89.3	90.4	89.8	-
	94.2	98.8	-	96.5
R2YOLOX-L(ours)	89.4	90.4	89.9	-
	96.0	98.9	-	97.5
R2YOLOX-X(ours)	89.7	90.5	**90.1**	-
	96.4	99.0	-	**97.7**

are visualized in Fig. 10. Besides, the R2YOLOX-L model not only achieves 79.12% mAP but also keeps the 33.8 FPS in inference, which is the best speed-accuracy trade-off model as we know. In addition, when using multiscale training and testing methods, our R2YOLOX-S model achieves 77.44% mAP and keeps the fastest inference at 58.2 FPS, this could expect to be deployed to actual UAV.

2) Results on HRSC2016: Note that the performance under VOC2012 metrics is often better than under VOC2007 metrics. DRN [16] and CenterMap [9] are only evaluated under pattern analysis, statistical modeling, and computational learning visual object classes (PASCAL VOC)2012 metrics, PIoU [37], RoI Transformer [7], Gliding Vertex [10], DAL [28] PolarDet [50], SKNet [32], and SKFat [33] are just evaluated under VOC2007 metrics, while other methods are evaluated under PASCAL VOC2007 and PASCAL VOC2012 metrics.

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TABLE VIII

COMPARISONS WITH SOTA RESULTS ON FGSD2021. THE SHORT NAMES FOR CATEGORIES ARE DEFINED AS (ABBREVIATION-FULL NAME): AIR-AIRCRAFT CARRIERS, WAS-WASP CLASS, TAR-TARAWA CLASS, AUS-AUSTIN CLASS, WHI-WHIDBEY ISLAND CLASS, SAN-SAN ANTONIO CLASS, NEW-NEWPORT CLASS, TIC-TICONDEROGA CLASS, BUR-ARLEIGH BURKE CLASS, PER-PERRY CLASS, LEW-LEWIS, CLARK CLASS, SUP-SUPPLY CLASS, KAI-HENRY J. KAISER CLASS, HOP-BOB HOPE CLASS, MER-MERCY CLASS, FRE-FREEDOM CLASS, IND-INDEPENDENCE CLASS, AVE-AVENGER CLASS, SUB-SUBMARINE AND OTH-OTHER. THE BEST RESULTS IN EACH CATEGORY ARE HIGHLIGHTED IN BOLD

Methods	Air	Was	Tar	Aus	Whi	San	New	Tic	Bur	Per	Lew	Sup	Kai	Hop	Mer	Fre	Ind	Ave	Sub	Oth	mAP (%)
R^2CNN [51]	89.9	80.9	80.5	79.4	87.0	87.8	44.2	89.0	89.6	79.5	**80.4**	47.7	81.5	87.4	**100**	82.4	**100**	66.4	50.9	57.2	78.1
RoI Transformer [7]	90.9	88.6	87.2	89.5	78.5	88.8	81.8	89.6	89.8	90.4	71.7	74.7	73.7	81.6	78.6	**100**	75.6	78.4	68.0	66.9	83.4
SCRDet [13]	77.3	90.4	87.4	89.8	78.8	90.9	54.5	88.3	89.6	74.9	68.4	59.2	**90.4**	77.2	81.8	73.9	**100**	43.9	43.8	57.1	75.9
R^3Det [27]	90.9	80.9	81.5	90.1	79.3	87.5	29.5	77.4	89.4	69.7	59.9	67.3	80.7	76.8	72.7	83.3	90.9	38.4	23.1	40.0	70.5
S^2ANet [11]	90.9	81.4	73.3	89.1	80.9	89.9	81.2	89.2	90.7	88.9	60.5	75.9	81.6	89.2	**100**	68.6	90.9	61.3	55.7	64.7	80.2
CHPDet [34]	90.9	90.9	90.6	90.5	90.3	90.3	**90.3**	89.9	90.2	90.2	78.2	**92.4**	88.9	89.2	**100**	99.4	99.4	81.9	88.6	57.0	88.9
R2YOLOX-S(ours)	99.9	93.1	91.0	93.9	89.6	97.9	83.2	96.1	96.7	91.1	57.6	87.0	86.6	**97.7**	10.0	80.1	97.9	88.6	68.0	75.6	88.6
R2YOLOX-L(ours)	**100**	93.2	91.3	**96.5**	**95.9**	97.3	87.2	96.2	97.6	95.8	70.0	81.1	86.7	85.2	84.6	90.2	99.7	90.1	84.6	78.0	90.1
R2YOLOX-X(ours)	**100**	**95.6**	**96.7**	94.7	95.4	**98.2**	82.6	**98.6**	**98.3**	**98.7**	67.6	89.7	88.0	97.3	99.5	90.1	99.2	**92.5**	**90.3**	**80.1**	**92.5**

Fig. 11. Examples of detection results on the HRSC2016 dataset using our proposed R2YOLOX-L. The results whose scores are higher than 0.3 are shown.

As shown in Table VI, our proposed method can achieve 90.1% and 97.7% mAP under VOC2007 and VOC2012 metrics, respectively, which can compete with other anchor-based methods. And as far as we know, our method achieves the SOTA performance on the VOC2012 metrics. These results prove that the anchor-free methods we proposed are already no weaker than the anchor-based method. Fig. 11 gives some examples of detection results on the HRSC2016 dataset.

3) Results on UCAS-AOD: The UCAS-AOD dataset contains a large number of small objects, which are often overwhelmed by complex surrounding scenes in aerial images. In order to comprehensively compare with other methods, we use VOC2007 and VOC2012 metrics to evaluate the performance. The experiment results are listed in Table VII, our method achieves the best mAP values of 97.7% and 90.1% mAP under VOC2012 and VOC2007 metrics, respectively. On the VOC2012 metric, we obtain SOTA performance, which proves the superiority of the performance of our proposed method. The visualization results of vehicle and airplane detection are shown in Fig. 12.

4) Results on FGSD2021 [34]: Since our methods have achieved high performance in detecting large aspect ratio objects such as ports, SHs, and BRs on the DOTA dataset, therefore, in this article, we select the FGSD2021 dataset to further explore the performance of our model in the task of large aspect ratio fine-grained detection. The performance comparison results between our method and some SOTA methods on the FGSD2021 dataset are shown in Table VIII. Our method achieves 92.5% mAP, which is the new SOTA

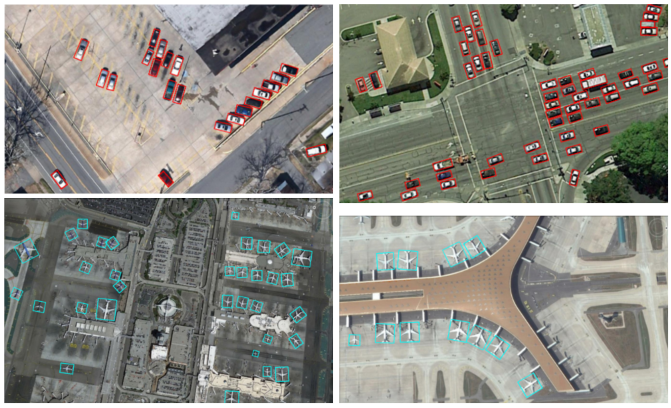

Fig. 12. Examples of detection results on the UCAS-AOD dataset using our proposed R2YOLOX-X.

Fig. 13. Examples of detection results on the FGSD2021 dataset using our proposed R2YOLOX-X.

performance, exceeding all other methods, and proves the superiority of our method. The visualization results of our proposed method are shown in Fig. 13.

V. DISCUSSION

Ablation experiments show that our proposed RRM and GSOTA methods can effectively improve the performance of the baseline. Combining RRM and GSOTA, we proposed R2YOLOX series models, which show the competitive performance results on four datasets, i.e., DOTA, HRSC2016, UCAS-AOD, and FGSD2021. However, the proposed method

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also has some problems. 1) Due to the following, the YOLOX framework, we use only one key point (i.e., the center point) to detect the target, sometimes it may be difficult to reflect all the characteristics of the target. 2) Because using AlignConv to align features, although it can achieve certain performance enhancements, it also introduces the difficulty of computational deformable convolution for embedded deployment. 3) Compared with other methods [11], our proposed R2YOLOX model needs more training epochs (40 epochs) to get model convergence, but those methods which usually use ResNet-50 as the backbone just need 12 training epochs. We think this is because we use the pretrained model trained on COCO, which is less universal and robust than those trained on ImageNet. However, like the original YOLOX which needs training 300 epochs from scratch, the training consume times are unacceptable (i.e., our device needs training for 20 days to get one model). Therefore, to balance time consumption and detection accuracy, in this article, we use the pretrained model on COCO.

In future work, we will try to use the multikey point architectures for reference to solve the problems of using a single key point, and the recent method [18] is a good example. For the AlignConv, we will try to replace it with other methods which are more lightweight and embedded systems-friendly methods. And the GSOTA method we proposed can also be used in many anchor-based and anchor-free methods, which need verification by further experiments.

VI. Conclusion

In this article, we proposed a lightweight refined anchor-free rotated detector for ODAIs, named R2YOLOX. We mainly add an RRM and use the Gaussian distribution Sampling Optima Transport Assignment method for training, which is more suitable for rotated detection tasks. We conduct extensive experiments on four challenging aerial object detection benchmarks. Experimental results show that our methods have competitive accuracy with the current advanced anchor-based two-stage and one-stage detectors while maintaining the best inference efficiency.

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